

Speak Like A Mathematician:

converse:

flow proof:

paragraph proof:

Transitive Property of Parallel Lines:

Converses — Proving Lines Are Parallel

Corresponding Angles Converse: if a pair of corresponding angles are congruent, then the lines are _____.

Alternate Interior Angles Converse: if a pair of alternate interior angles are _____, then the lines are parallel.

Alternate Exterior Angles Converse: if a pair of alternate exterior angles are _____, then the lines are parallel.

Consecutive Interior Angles Converse: if a pair of consecutive interior angles are _____, then the lines are parallel.

Example 1 - Choose the Word

Choose the word that completes each statement correctly.

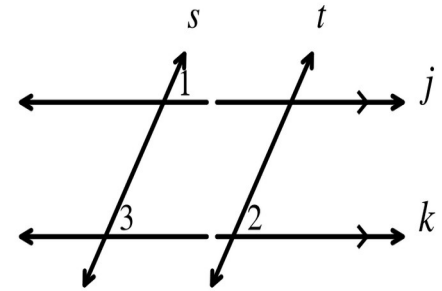
- a) If two lines are cut by a transversal so the alternate interior angles are (congruent / supplementary), then the lines are parallel.
- b) If two lines are cut by a transversal so the consecutive interior angles are (congruent / supplementary), then the lines are parallel.
- c) If two lines are cut by a transversal so the corresponding angles are (congruent / supplementary), then the lines are parallel.

Example 2 - Proving Two Lines Are Parallel

Given: $j \parallel k$, $\angle 1 \cong \angle 2$

Prove: $s \parallel t$

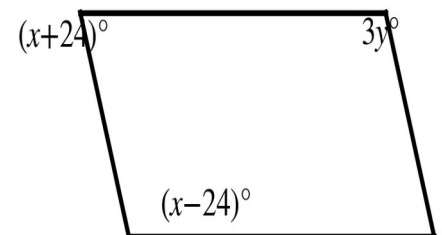
Statements	Reasons
1. $j \parallel k$	1. Given
2. $\angle 1 \cong \angle 2$	2.
3.	3.
4.	4.
5.	5.



Example 3 - Find x and y

Find the values of x and y that make the opposite sides of the figure parallel.

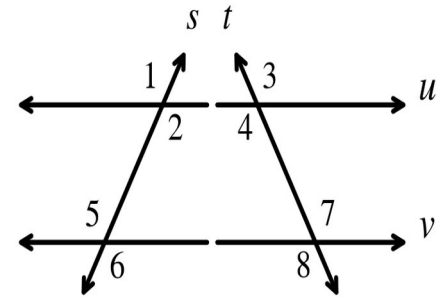
- Use the two angles on the left side.
- Use the two angles along the top.
- State the values.



Example 4 - Which Lines Are Parallel?

Use the diagram. Which two lines (if any) must be parallel? Name the theorem that justifies it.

- 1) $\angle 1 \cong \angle 5$
- 2) $\angle 2 \cong \angle 5$
- 3) $\angle 3 \cong \angle 7$
- 4) $\angle 2$ and $\angle 4$ are supplementary
- 5) $\angle 1 \cong \angle 6$
- 6) $\angle 4 \cong \angle 7$
- 7) $\angle 3$ and $\angle 7$ are supplementary



Example 5 - Show That Two Lines Are Parallel

In the diagram, $m\angle 2 = (3x - 14)^\circ$ and $m\angle 6 = (2x + 11)^\circ$, and $x = 25$. Show that $r \parallel s$.

- a) Find $m\angle 2$ and $m\angle 6$.
- b) Explain why $r \parallel s$.

